



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

## GKE PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Cloud

TOTAL: 10 QUESTIONS

**Q1** What is GKE and what is its primary purpose in its cloud ecosystem?

Threat Model

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**Q6** How do you manage configuration and secrets for GKE?

Observability

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**Q2** How does IAM work with GKE?

Architecture

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**Q7** What are the main cost drivers for GKE and how do you optimize them?

Lifecycle

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**Q3** How do you monitor GKE in production?

Configuration

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**Q8** How do you implement high availability with GKE?

Compliance

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**Q4** How do you scale GKE to handle variable load?

Hardening

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**Q9** How does GKE integrate with a CI/CD pipeline?

Testing

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**Q5** How do you secure GKE in production?

SSO / IdP

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**Q10** What are common pitfalls when first adopting GKE?

Tuning

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 GKE is a managed cloud service

Mitigation

GKE is a managed cloud service that handles a specific infrastructure concern (compute, storage, messaging, AI). Using it shifts operational burden to the cloud provider while you focus on application logic.

A6 Store sensitive values in a managed

Observability

Store sensitive values in a managed secret store (AWS Secrets Manager, GCP Secret Manager, Azure Key Vault). Inject secrets as environment variables at runtime, never bake them into container images or source code.

A2 Access to GKE is controlled via

Primitives

Access to GKE is controlled via IAM roles and policies. Attach the minimum required permissions to a service account or role. Avoid long-lived access keys; use instance roles or Workload Identity Federation instead.

A7 Cost in GKE typically comes from

Automation

Cost in GKE typically comes from compute hours, data transfer, storage, and request counts. Right-size instances, use reserved capacity for steady-state workloads, and enable lifecycle policies to expire old data.

A3 GKE emits metrics to the cloud

Hardening

GKE emits metrics to the cloud provider's monitoring service (CloudWatch, Cloud Monitoring, Azure Monitor). Set alerts on key metrics (error rate, latency, capacity). Use distributed tracing to correlate across services.

A8 Deploy GKE across multiple availability zones.

Governance

Deploy GKE across multiple availability zones. Use managed replicas or active-active configurations. Implement health checks and automatic failover. Test resilience regularly with chaos engineering or failover drills.

A4 GKE supports auto-scaling policies based on

Gotchas

GKE supports auto-scaling policies based on CPU, queue depth, or custom metrics. Set minimum and maximum capacity. Horizontal scaling (more instances) is generally preferred over vertical for cloud workloads.

A9 CI/CD pipelines use the cloud CLI

Assessment

CI/CD pipelines use the cloud CLI or SDK to deploy updates to GKE after tests pass. Infrastructure-as-code (Terraform, CDK) defines GKE configuration as version-controlled code deployed via the pipeline.

A5 Place GKE in a private subnet

Federation

Place GKE in a private subnet with no public endpoint where possible. Use VPC security groups or firewall rules to allow only required traffic. Encrypt data at rest and in transit. Rotate credentials regularly.

A10 Common mistakes include misconfigured IAM

Optimization

Common mistakes include misconfigured IAM policies (too permissive), no cost alerting, deploying only in one availability zone, and missing backups. Read the service SLA carefully to understand what the cloud provider guarantees.